EXACT VALUES AND STRUCTURAL PROPERTIES OF THE CYCLE–EDGE DECOMPOSITION NUMBER $\mathrm{ce}(G)$

THE FRONT184 COLLABORATION

Abstract. A *decomposition* of a graph G is a partition of $E(G)$ into edge-disjoint cycles and single edges, and $\mathrm{ce}(G)$ denotes the minimum possible number of parts. Erdős and Gallai proved in 1959 that $O(n \log n)$ parts always suffice and conjectured that $O(n)$ do; the conjecture is still open, the best bound being the $O(n \log^* n)$ of Bucić–Montgomery, while the sharper suggestion that $n-1$ parts might suffice fails already on $K_{3,n-3}$. This paper maps where the $n-1$ budget holds exactly and where it breaks. We determine $\mathrm{ce}(K_n)$ for every n; we prove that ce is exactly additive over blocks, so that cut-vertices never help, and derive a transfer principle that reduces every linear bound to its 2-connected case. For bounded degree, the $n-1$ bound fails at maximum degree six—the graph formed by a triangle and a $K_{3,4}$ on the same three vertices has $n = 7$ and $\mathrm{ce} = 7$—while for Eulerian graphs of maximum degree at most six we prove $\mathrm{ce}(G) \le n-1$ without exception. The latter resolves the six-regular case of a problem of Akbari–Aloni–Beikmohammadi–Clow, with a strengthened bound: every 6-regular graph on n vertices with $n \equiv 0 \pmod 3$ satisfies $\mathrm{ce}(G) \le n-2$. The proof rests on a rigidity lemma, which says that a 4-regular graph whose 2-factors are all unions of triangles must be the line graph $L(B)$ of a cubic bipartite graph, together with an explicit factorisation of such line graphs into two triangle-free 2-factors (an explicit form of an observation of Kouider–Sabidussi). We also prove the cubic case of the same authors' odd-regular conjecture for the relaxed invariant $\mathrm{f_{re}}$, reduce the general $\Delta \le 6$ question to an explicitly identified T-join danger zone, and establish a sharp degeneracy threshold: $\mathrm{ce} \le n-1$ holds for every 2-degenerate graph and fails at every degeneracy $d \ge 3$.

Contents

1. Introduction 2
1.1. The problem 2
1.2. This paper 2
1.3. Relation to known results 3
2. Preliminaries 3
3. Complete graphs and the block calculus 4
4. Planar, toroidal, and positive-genus families 6
5. Dirac-regular graphs 7
6. Maximum degree six: the counterexample and the six-regular case 8
6.1. The counterexample 8
6.2. Eulerian graphs of maximum degree six 8
6.3. The general $\Delta \le 6$ question 11
7. Odd degrees: the cubic case of Conjecture 6.4 11
8. Bounded degeneracy 14
9. Extremal examples and the shape of $f(n)$ 15
10. Anatomy of the $\log^*$ barrier 16
11. Concluding remarks and open problems 17
Appendix A. Proof status and computational verification 18
General results 18

Draft v2, September 16, 2026. Every statement below has passed two independent adversarial review rounds, and all computational claims are summarised in Appendix A. Proofs given here are sketches; the complete proofs are maintained in the project records and are referenced per theorem.

Six-regular graphs 19
Cubic and odd-regular graphs 19
Acknowledgments 20
References 20

1. Introduction

1.1. **The problem.** Let G be a finite simple graph on n vertices and m edges. A *part* is either a single edge or the edge set of a simple cycle, and a *decomposition* of G is a partition of $E(G)$ into parts. The minimum number of parts is $\mathrm{ce}(G)$, the cycle–edge decomposition number of G. The definition is naive; the function is not.

Erdős and Gallai [14] proved that every n-vertex graph admits a decomposition into $O(n \log n)$ parts, and conjectured that $O(n)$ always suffice; the conjecture in this form is recorded in [15] and is listed as Problem #184 in Erdős' problem collection [16]. Erdős also suggested, in [17], that $n-1$ parts might always be enough. Read without edge-disjointness, that suggestion is a theorem: Pyber [23] showed that $n-1$ cycles and edges always *cover* the edges of a graph. Read with edge-disjointness, it is false, and it fails early: the complete bipartite graph $K_{3,n-3}$ already forces $(4/3)(n-3)$ parts, and the construction of Erdős [13], which uses the complete bipartite graphs $K_{2k+1,\,n-2k-1}$ and generalises an example of Gallai, forces $(3/2-o(1))n$ parts; both computations are carried out in [6]. And yet no graph is known to need more than $(3/2-o(1))n$ parts. The linear story is wide open. After Conlon–Fox–Sudakov [8] improved the general upper bound to $O(n \log\log n)$, Bucić and Montgomery [6] obtained the current best, $O(n \log^* n)$.

Why should one study the precise question $n-1$, given that it is false? Because the counterexamples are highly structured, and knowing exactly which families meet the budget and which break it says something about where the true constant of the linear conjecture comes from. That is the point of view of this paper: we treat $n-1$ as a budget and audit it, family by family.

1.2. **This paper.** The audit proceeds on four fronts. First, structure: cut-vertices never help. We prove that ce is exactly additive over blocks, $\mathrm{ce}(G) = \sum_B \mathrm{ce}(B)$ (Theorem 3.2), so the entire difficulty of the problem lives on 2-connected blocks, and every linear bound can be transferred between general graphs and their blocks (Corollary 3.3). At the two ends of the density range the budget is met with zero slack, for opposite reasons: trees have no cycles to merge, while even-order complete graphs are pinned by parity — every decomposition of K_{2m} needs at least m single edges — and the two accounts close at exactly $n-1$ (Theorem 3.1).

Second, bounded degree. The $n-1$ bound holds for $\Delta \le 4$ [1], whose authors ask what happens at $\Delta = 5$ and $\Delta = 6$. At $\Delta = 6$ the budget breaks: the graph $K_3 \cup K_{3,4}$ — a $K_{3,4}$ and a triangle on the same three vertices — has $n = 7$, $\Delta = 6$, and $\mathrm{ce} = 7 = n$ (Theorem 6.1). The interesting news is on the Eulerian side. For Eulerian graphs of maximum degree 6 we close the problem completely at $n-1$, and this resolves the six-regular case of Problem 6.1 of [1] with a strengthened bound

$$\text{every 6-regular simple graph with } n \equiv 0 \pmod 3 \text{ satisfies } \mathrm{ce}(G) \le n-2,$$

together with the verbatim proof of Conjecture 6.2 of [1] (every 6-regular graph decomposes into three 2-factors, one of which has a component of order at least 4). The obstacle that had held up the six-regular case is genuinely structural. If all 2-factors of a 4-regular graph are unions of triangles, then the graph is forced to be the line graph $L(B)$ of a cubic bipartite graph (Lemma 6.5); on such line graphs, two complementary 2-factors $\Phi(M)$ and $\Phi(-M)$, built from an orientation of a perfect matching of B, partition the edge set into two triangle-free 2-factors (Proposition 6.6) — a contradiction. That last construction is an explicit, self-contained special case of an observation of Kouider–Sabidussi [20], and we claim no priority for the existence statement itself; the reduction, the explicit partition, and the resulting

bounds are, to the best of our knowledge, new. The general $\Delta \le 6$ question is then reduced to a T-join bound with an honestly identified danger zone (Theorem 6.12, Remark 6.13).

Third, odd degrees on the relaxed invariant. The paper [1] also studies $\mathrm{f_{re}}(G)$, the minimum number of blocks in a partition of $E(G)$ into 2-regular subgraphs and single edges, and conjectures (Conjecture 6.4 there) that every $(2k+1)$-regular graph has $\mathrm{f_{re}} \le n-1$. The case of graphs with a perfect matching is easy and was already observed in [1]; the open core is the graphs without one. We prove the cubic case unconditionally: every cubic graph satisfies $\mathrm{f_{re}}(G) \le n-1$ (Theorem 7.6). The argument is a reduction chain — an exact formula $\mathrm{f_{re}} = p+1$ on cubic graphs in terms of the minimum T-join, a decomposition of p over the block tree, and a lower bound on even subgraphs of bridgeless subcubic blocks — whose last step is quoted from the published Theorem 1.3 of Choi–Kim–Kostochka–Park–West [9]; our contribution is the chain itself and an exhaustive computational verification.

Fourth, sparsity. The degeneracy hierarchy collapses cleanly: $\mathrm{ce} \le n-1$ holds for every 2-degenerate graph — equivalently every K_4-minor-free graph — and fails, by arbitrarily large margins, at every degeneracy $d \ge 3$ (Theorems 8.1 and 8.6). The extremal family is the complete bipartite graph $K_{3,q}$, for which we determine ce exactly for all q (Theorem 8.3), so the asymptotic constant of the 3-degenerate class is pinned to $[4/3, 5/3]$.

Finally, we dissect the Bucić–Montgomery architecture from the inside (Section 10): the polylogarithmic remainder map is locked to depth $\log^* n$ by a superconstant fixed point; the apparently better remainder $2^{(\log d)^\theta}$ in fact yields depth $\Theta(\log\log\log n)$, which is *worse*; and no amount of edge deletion can create the expanders the method needs. These are architecture-relative statements — they locate what any improvement must change, not unconditional lower bounds against all algorithms.

1.3. **Relation to known results.** Pyber's covering theorem [23] gives $n-1$ parts when edges may be reused; our block calculus shows that for the decomposition version nothing is ever gained or lost at a cut-vertex, so the whole difficulty is 2-connected. The planar case of the decomposition problem does not appear to have been studied before; a literature search in September 2026 found no prior work, though we have not re-verified this against the full literature. The toroidal and positive-genus bounds of Section 4 appear to be new, with the same caveat, as does the K_4-minor-free statement of Theorem 8.1 (flagged as an unverified-novelty item in our own records). For Theorem 9.1 no novelty is claimed at all: the lower bounds are known, the computation is the same as in [6, 13], and we restate it only because our extremal families are built on it. The six-regular factorisation of line graphs into triangle-free 2-factors (Proposition 6.6) is an explicit special case of Kouider–Sabidussi [20]; the related algorithmic result of Bertram–Horák [4] decides whether a 4-regular graph decomposes into two triangle-free 2-factors but contains no counting consequence for ce. Where our extremal mechanisms coincide with the known $(3/2 - o(1))n$ construction — parity-forced single edges at the dense side of an odd complete bipartite kernel — we say so in place.

2. Preliminaries

All graphs are finite and simple. For a graph G we write $n = |V(G)|$, $m = |E(G)|$, $\Delta = \Delta(G)$, and $c(G)$ for the number of connected components. A *part* is a single edge or the edge set of a simple cycle of length at least 3.

Definition 2.1. A *decomposition* of G is a partition of $E(G)$ into parts. $\mathrm{ce}(G)$ is the minimum number of parts in a decomposition, with $\mathrm{ce}(G) = 0$ when $E(G) = \emptyset$. Let $f(n) = \max\{\mathrm{ce}(G) : |V(G)| = n\}$.

Two identities do most of the elementary work. The first converts decompositions into cycle packing; the second converts parity into forced single edges.

Lemma 2.2 (Budget/savings identity)**.** *For any decomposition with k cycles of lengths $\ell_1, \dots, \ell_k$ and s single edges, $k + s = m - \sum_i (\ell_i - 1)$. Consequently*

$$\mathrm{ce}(G) \;=\; m - \max_{\mathcal{C}} \; \textstyle\sum_{C \in \mathcal{C}} (|C| - 1),$$

where the maximum is over edge-disjoint cycle families $\mathcal{C}$ (all remaining edges are taken as single-edge parts).

Proof status: Complete proof in campaign records, `campaign/lane_03_constant.md` *§0 and* `campaign/lane_04_delta5_savings.md` *§1; elementary..*

Definition 2.3. Let $T = T(G)$ be the set of odd-degree vertices. A subgraph $F \subseteq G$ is a *parity subgraph* if $\deg_F(v) \equiv \deg_G(v) \pmod 2$ for all v; equivalently F is a T-join in the sense of Edmonds–Johnson [12]. Let $\tau(G)$ be the minimum number of edges of a parity subgraph, and let $\mathrm{odd}(G) = |T|$.

Lemma 2.4 (Parity bounds)**.** *In any decomposition the single edges form a parity subgraph; hence $2s \geq \mathrm{odd}(G)$ and $s \geq \tau(G)$ for the number s of single edges, and $\mathrm{ce}(G) \geq \mathrm{odd}(G)/2$.*

Proof status: Complete proof in campaign records, `campaign/lane_09_blocks.md` *Lemma F; elementary double counting..*

Lemma 2.5 (T-join upper bound)**.** *For every connected graph G, $\mathrm{ce}(G) \leq \tau(G) + \lfloor (m - \tau(G))/3 \rfloor \leq \lfloor (m + 2\tau(G))/3 \rfloor$. Consequently, if $m + 2\tau(G) \leq 3n - 3$ then $\mathrm{ce}(G) \leq n - 1$; and $\tau(G) \leq n - 1$ always, via a spanning tree, so the criterion applies to every family with m bounded by $3n + O(1)$.*

Proof sketch. Take a minimum parity subgraph F with $|F| = \tau$. Then $G - F$ is Eulerian, hence decomposes into at most $\lfloor (m - \tau)/3 \rfloor$ edge-disjoint cycles; add the τ edges of F as single-edge parts. The criterion is integer arithmetic. For the last claim, the unique minimal T-join inside a spanning tree has at most $n - 1$ edges. □

Proof status: Complete proof in campaign records, `campaign/lane_04_delta5_savings.md` *Thm. 1.2 ("R1") and* `campaign/lane_08_degeneracy.md` *Lemmas 3.1–3.2 (independent re-derivation). Note that the bound is meaningful only for families with $m = O(n)$; used with $m \leq dn - \binom{d+1}{2}$ (degenerate graphs, Theorem 8.2) and with $m \leq 3n$ (maximum degree six, Theorem 6.12)..*

Definition 2.6. A *block* of G is a maximal subgraph with no cut-vertex of itself; every block is a bridge K_2 or 2-connected, the blocks partition $E(G)$, and any two blocks share at most one vertex. The *degeneracy* of G is $d = \max_{H \subseteq G} \delta(H)$, and the d-core is the unique maximal subgraph of minimum degree at least d. The *orientable genus* $g(G)$ is the minimum genus of an orientable surface in which G embeds; all embeddings below are cellular. Orientable genus is additive over blocks [3], whereas non-orientable genus is not [25]; all genus statements in this paper are therefore for the orientable invariant.

3. Complete graphs and the block calculus

Theorem 3.1 (Complete graphs)**.** *For all $m \geq 2$, $\mathrm{ce}(K_{2m}) = 2m - 1$; for all $m \geq 1$, $\mathrm{ce}(K_{2m+1}) = m$.*

Proof sketch. The upper bounds are the classical Walecki constructions: K_{2k+1} decomposes into k edge-disjoint Hamilton cycles, and K_{2k} into $k - 1$ Hamilton cycles plus a perfect matching; see [21] and [5]. For the lower bounds, let $n = 2k + 1$: every part has at most n edges, so $\mathrm{ce} \geq \lceil m/n \rceil = (n-1)/2$. Let $n = 2k$: all $2k$ vertices are odd, so any decomposition contains $s \geq k$ single edges by Lemma 2.4; with k' cycles, $k' \geq (m-s)/(2k)$ and $\mathrm{ce} = s + k' \geq (2k-1)/2 + s(2k-1)/(2k) \geq 2k - 1 = n - 1$. □

Proof status: Complete proof in campaign records, `campaign/lane_09_blocks.md` *Thm. G (self-contained lower bounds; designated proof of record). Numerics:* $\mathrm{ce}(K_4) = 3$, $\mathrm{ce}(K_5) = 2$, $\mathrm{ce}(K_6) = 5$, $\mathrm{ce}(K_7) = 3$, $\mathrm{ce}(K_8) = 7$ *by independent exact solvers, that for* K_8 *being a* 3.3×10^7*-step search log; see Appendix A..*

The exact values show two opposite ways of sitting exactly at the budget. Trees do it for lack of material; K_{2m} does it at full density, because parity pins down $n/2$ single edges and the accounting then closes at exactly $n-1$. Since zero-slack 2-connected blocks of the second kind exist at every even order, no uniform per-block discount can exist. The next results make this precise.

Theorem 3.2 (Block additivity)**.** *For every graph* G *with block set* $\mathcal{B}(G)$*,*

$$\mathrm{ce}(G) \;=\; \sum_{B \in \mathcal{B}(G)} \mathrm{ce}(B).$$

Moreover every connected graph satisfies the counting identity $\sum_B (|V(B)| - 1) = n - 1$*, and more generally* $\sum_B (|V(B)| - 1) = n - c(G)$*.*

Proof sketch. For ($\leq$), concatenate optimal decompositions of the blocks; block edge sets are disjoint. For ($\geq$), observe that every part is 2-regular-connected or a single edge, hence lies in a unique block — a cycle meeting two blocks would force two blocks to share three vertices. Grouping parts by block therefore yields decompositions of the blocks, so the number of parts is at least the right-hand side. The counting identity follows by induction on the block–cutvertex tree. □

Proof status: Complete proof in campaign records, `campaign/lane_09_blocks.md` *Thm. A and Lemma 1.6 (which includes a recorded instance where a* 300*-instance randomized check corrected the identity's first draft)..*

Block additivity is an equality, not an inequality: no decomposition can ever exploit a cut-vertex to save a part. Two corollaries quantify this.

Corollary 3.3 (Transfer principle)**.** *For* $\alpha \geq 1$*: every* 2*-connected graph* H *satisfies* $\mathrm{ce}(H) \leq \alpha(|V(H)| - 1)$ *if and only if every graph* G *satisfies* $\mathrm{ce}(G) \leq \alpha(n - c(G))$*. In particular, for* $\alpha = 1$ *the* $n - 1$ *problem is exactly equivalent to its* 2*-connected restriction.*

Corollary 3.4 (Excess additivity)**.** $\mathrm{ce}(G) - (n - c(G)) = \sum_B \big[\mathrm{ce}(B) - (|V(B)| - 1)\big]$*: the excess over the linear budget is the sum of per-block excesses, with no cancellation between blocks.*

Proof status: Complete proofs of Corollaries 3.3–3.4 in campaign records, `campaign/lane_09_blocks.md` *Cor. B/C..*

Proposition 3.5 (Merge criterion)**.** *(i) In any decomposition, two distinct parts can never be merged into one part. (ii)* $k \geq 2$ *parts can be merged into one part if and only if the union of their edge sets is the edge set of a single cycle. Consequently the cut-vertex discount is exactly zero: no restructuring across, or at, a cut-vertex ever saves a part.*

Proof sketch. A union of two parts is the edge set of a cycle only if it is connected and 2-regular. Two cycles sharing a vertex give degree 4 there, and disjoint ones are disconnected; a cycle plus a single edge gives degree 3 at an endpoint; two single edges are not 2-regular. □

Proof status: Complete proof in campaign records, `campaign/lane_09_blocks.md` *Thm. D (minimal witness: the bowtie,* $\mathrm{ce} = 2$*)..*

Theorem 3.6 (Cactus graphs)**.** *Let* G *be connected with every block a bridge or a cycle, and let* t *be the number of blocks. Then* $\mathrm{ce}(G) = t = (n - 1) - \sum_{C \text{ cycle block}} (|C| - 2)$*.*

Proof status: Complete proof in campaign records, `campaign/lane_09_blocks.md` *Thm. E..*

Remark 3.7. The two zero-slack mechanisms push in opposite directions, and together they explain why the $n-1$ problem resists both sparse and dense intuition. Numerically, $\mathrm{ce}(K_{2m}) = 2m-1$ is attained at every tested even order up to 8, while $\mathrm{ce}(K_{2m+1}) = m \approx n/2$: odd complete graphs enjoy a factor-2 discount, their even cousins none at all.

4. Planar, toroidal, and positive-genus families

Theorem 4.1 (Outerplanar graphs)**.** *Every n-vertex outerplanar graph satisfies $\mathrm{ce}(G) \le n-1$.*

Proof sketch. Reduce to blocks. A 2-connected outerplanar graph has a Hamilton outer cycle, and its weak dual is a tree. Peel a leaf face: its boundary is one chord xy plus an outer arc whose internal vertices all have degree exactly 2, so the arc together with xy forms a cycle C^* whose deletion leaves an outerplanar graph. Induction gives $\mathrm{ce}(G) \le 1 + \mathrm{ce}(G') \le 1 + (n-k) - 1 \le n-1$, where k is the number of peeled internal vertices. The degenerate one-chord case is handled explicitly in the archived proof. □

Proof status: Complete proof in campaign records, `notes/attack_planar.md` *Thm. A with Lemmas A1–A3; the proof underwent an adversarial fix round (degree-2 property of arc-internal vertices; degenerate case)..*

Theorem 4.2 (Eulerian plane graphs)**.** *Every $n \ge 3$-vertex Eulerian plane graph satisfies $\mathrm{ce}(G) \le n-2$; in fact $E(G)$ decomposes into at most $n-2$ cycles, with no single edges.*

Proof sketch. An Eulerian graph decomposes into edge-disjoint cycles, each of length at least 3, and $m \le 3n-6$. □

Proof status: Complete proof in campaign records, `notes/attack_planar.md` *Thm. B..*

Theorem 4.3 (Eulerian maximal plane graphs)**.** *If G is an Eulerian maximal plane graph, then $E(G)$ decomposes into exactly $n-2$ edge-disjoint facial triangles. This is an existence statement for a structured decomposition; $n-2$ need not be $\mathrm{ce}(G)$ — the octahedron has $\mathrm{ce} = 2$.*

Proof sketch. The dual G^* is 3-regular and bipartite, since a plane graph is Eulerian exactly when its dual is bipartite. A bipartition class of G^* has $(2n-4)/2 = n-2$ faces, and every edge of G lies in exactly one of them. □

Proof status: Complete proof in campaign records, `notes/attack_planar.md` *Thm. C (the "minimality" claim of a first draft was withdrawn in review)..*

Theorem 4.4 (Bridgeless cubic graphs)**.** *Every n-vertex bridgeless cubic graph satisfies $\mathrm{ce}(G) \le 5n/6$, which is $\le n-1$ for $n \ge 6$. Cubic graphs with bridges satisfy $\mathrm{ce}(G) \le n-1$.*

Proof sketch. By the Petersen theorem [22] a bridgeless cubic graph has a perfect matching M; then $G - M$ is a 2-factor with at most $n/3$ cycles. Decompose into M as single edges plus those cycles. Planarity is not used. The bridged case follows from $\Delta \le 4$ [1]. □

Proof status: Complete proof in campaign records, `notes/attack_planar.md` *Thm. D/D′ and* `campaign/lane_06_genus.md` *L6.4..*

Theorem 4.5 (Eulerian toroidal graphs)**.** *Every Eulerian graph embeddable in the torus satisfies $\mathrm{ce}(G) \le n-1$; moreover any triangulation component satisfies $\mathrm{ce} \le n-2$.*

Proof sketch. Work per component. If $e \le 3n-1$, the Euler decomposition gives at most $\lfloor e/3 \rfloor \le n-1$ cycles. If $e = 3n$, the component is a torus triangulation, hence simple, 2-connected, and of even minimum degree at least 4; by Dirac's circumference theorem [11] it contains a cycle of length $\ge \min\{n, 2\delta\} \ge 7$, and removing it leaves an Eulerian graph decomposable into at most $\lfloor (3n_i - 7)/3 \rfloor = n_i - 3$ cycles, for a total of at most $n_i - 2$. □

Proof status: Complete proof in campaign records, `campaign/lane_06_genus.md` L6.3. The $g = 0$ specialisation gives only the weaker $n - 1$, so this theorem is compatible with, and does not subsume, the open planar conjecture $\mathrm{ce} \le n - 1$.*.*

Theorem 4.6 (Explicit genus bounds)**.** *Let G have orientable genus g. (i)* $\mathrm{ce}(G) \le (m + 2n - 2)/3 \le 5n/3 + 2g - 8/3$. *(ii) If moreover G is Eulerian, then* $\mathrm{ce}(G) \le \lfloor m/3 \rfloor$*; in particular* $\mathrm{ce}(G) \le n + 2g - 3$*, and* $\mathrm{ce}(G) \le n + 2g - 4$ *if G triangulates the genus-g surface.*

Proof sketch. (i) Greedily peel edge-disjoint cycles until a forest R remains; peeling k cycles of total length L costs $k + |E(R)| \le L/3 + |E(R)| \le m/3 + 2(n-1)/3$, and $m \le 3n - 6 + 6g$ by Euler's formula. (ii) The Euler decomposition has at most $\lfloor m/3 \rfloor$ cycles; the triangulation case gains one via the Dirac-cycle argument of Theorem 4.5. □

Proof status: Complete proof in campaign records, `campaign/lane_06_genus.md` L6.1/L6.2. Part (i) is a folkloristic observation recorded for completeness..

Corollary 4.7 (Eulerian plane triangulations)**.** *For $n \ge 8$, every Eulerian plane triangulation satisfies* $\mathrm{ce}(G) \le n - 4$*. No Eulerian plane triangulation exists for $n = 7$: its degree sequence would be $(6^1, 4^6)$, giving a universal vertex whose link is a 6-cycle with the remaining three edges drawn outside as a non-crossing perfect matching, that is, a 3-regular outerplanar graph — contradicting the fact that every outerplanar graph has two vertices of degree at most 2.*

Proof status: Complete proof in campaign records, `campaign/lane_06_genus.md` L6.2′ as revised in the review round; an earlier "spoke counting" argument there contained a counting error and was replaced by the degree-sequence argument. Exhaustive enumeration double-checks $n \le 7$, with the octahedron unique at $n = 6$ (Appendix A)..

Corollary 4.8 (Hamiltonian graphs of genus g)**.** *Let G have orientable genus g and contain a Hamilton cycle. Then* $\mathrm{ce}(G) \le 1 + (m-n)/3 + 2(n-1)/3 \le 4n/3 + 2g - 5/3$*. In particular, every 5-connected toroidal graph — which is Hamiltonian by Thomas–Yu [26] — satisfies* $\mathrm{ce}(G) \le 4n/3 + 1/3$.

Proof sketch. Delete the Hamilton cycle and apply Theorem 4.6(i) to the remainder; the last inequality uses $m \le 3n - 6 + 6g$. □

Proof status: Complete proof in campaign records, `campaign/lane_06_genus.md` L6.5..

Corollary 4.9 (A necessary growth rate in g)**.** *Any universal bound of the form "every genus-g graph has* $\mathrm{ce} \le n - 1 + f(g)$*" requires $f(g) \ge \lceil (4g - 4)/3 \rceil$ for $g \ge 2$: the graph $K_{3,\,4g+2}$ has genus exactly g by Ringel's formula [24], has $n = 4g + 5$, and satisfies* $\mathrm{ce}(K_{3,4g+2}) = \lceil (16g + 8)/3 \rceil$ *by Theorem 8.3.*

Proof status: Complete proof in campaign records, `campaign/lane_06_genus.md` L6.6 (assembled from Theorem 8.3, verified for all q, plus Ringel's formula, whose genus-2 window was checked exactly)..

5. Dirac-regular graphs

Theorem 5.1 (Dirac-regular graphs)**.** *Let n_0 be the constant of the Hamilton decomposition theorem of Csaba–Kühn–Lo–Osthus–Treglown [10], and let G be a D-regular graph on $n \ge n_0$ vertices with $D \ge \lfloor n/2 \rfloor$. Then:*

(i) *if n is odd,* $\mathrm{ce}(G) = D/2 \le (n-1)/2$*, with no single edges;*
(ii) *if n and D are even,* $\mathrm{ce}(G) = D/2 \le (n-2)/2$*, with no single edges;*
(iii) *if n is even and D is odd,* $\mathrm{ce}(G) = (D-1)/2 + n/2 \le n - 1$.

In particular every Dirac-regular graph on $n \ge n_0$ vertices satisfies $\mathrm{ce}(G) \le n - 1$.

Proof sketch. By [10], G decomposes into Hamilton cycles plus at most one perfect matching. Compare edge counts modulo n: $e(G) = Dn/2$ is divisible by n exactly when D is even, which forces the matching to be empty in cases (i)–(ii) (in case (i) no perfect matching exists at all) and forces it present with $t = (D-1)/2$ cycles in case (iii). Maximal D in each regime gives the displayed inequalities; (iii) at $D = n-1$ with n even gives exactly $n-1$. □

Proof status: Complete proof in campaign records, `campaign/lane_07_mindegree.md` *Thm. A, checked against the local copy of the source of* [10]*..*

Remark 5.2 (Scope). The restriction $n \geq n_0$ is essential and must not be dropped: n_0 is an astronomical constant, and the small-n, $D \in \{5, 6\}$ regular range is not covered by Theorem 5.1. No contradiction with the open status of $\Delta \leq 5$ follows — but neither is that case subsumed. The gap between Theorem 5.1 and its own lower-bound anchor ($\mathrm{ce}(K_n) \geq n/2$ for even n) is a factor of about 2; see Section 11 for the non-regular Dirac problem.

6. Maximum degree six: the counterexample and the six-regular case

The $n-1$ budget survives $\Delta \leq 4$ [1] and, as far as anyone can tell, $\Delta \leq 5$. This section is about where it dies, and about how much of the $\Delta \leq 6$ world can be saved.

6.1. **The counterexample.**

Theorem 6.1 ($\Delta \leq 6$ with ce $= n$)**.** *Let G have vertices $a_1, a_2, a_3, b_1, \ldots, b_4$ and edge set consisting of the triangle a_1a_2, a_2a_3, a_3a_1 plus all twelve edges a_ib_j; equivalently, G is the union of K_3 and $K_{3,4}$ sharing the vertex set $\{a_1, a_2, a_3\}$. Then $n = 7$, $m = 15$, $\Delta = 6$, and $\mathrm{ce}(G) = 7 = n$.*

Proof sketch. An explicit 7-part certificate — three cycles of lengths $4, 3, 4$ and four single edges — covers each of the 15 edges exactly once, so ce ≤ 7. For the lower bound, each b_j has odd degree 3, so at least one single edge is incident to each b_j (Lemma 2.4); any decomposition with at most 6 parts therefore has $c + s \leq 6$ with $s \geq 4$, leaving the cases $(c, s) \in \{(0,4), (0,5), (0,6), (1,4), (1,5), (2,4)\}$. The cases with $c \leq 1$ cannot cover 15 edges with cycles of length at most 7. For $c = 2, s = 4$: exactly one single edge is taken at every b_j, so the two cycles must supply all eight remaining edge-ends at the b-vertices. The lengths must be $\{4, 7\}$ or $\{5, 6\}$. A 7-cycle visits seven distinct vertices, hence all four b_j's, demanding eight b-side ends on seven edges — impossible. For $\{5, 6\}$: a 6-cycle is purely bipartite and touches three b_j's, while the 5-cycle, which then contains a triangle edge and only three a-vertices are available, needs two *untouched* b_j's — of which only one remains. □

Proof status: Complete proof in campaign records, `campaign/lane_05_delta6.md` *Thm. B. Verified four independent ways: a DP solver, branch-and-bound, a feasibility enumeration at $k = 5, 6$, and an independently written solver with zero shared code (*`campaign/prepub_independent_check.py`*), plus per-edge certificate checking (Appendix A)..*

Corollary 6.2. *"$\Delta \leq 6 \Rightarrow \mathrm{ce} \leq n - 1$" is false. Trees show that $n-1$ is attained already at $\Delta \leq 1$, and Theorem 6.1 shows that n is attained at $\Delta = 6$; so within $\Delta \leq 6$ the 0-slack behaviour of trees persists at the top degree, and n is the natural next ceiling.*

6.2. **Eulerian graphs of maximum degree six.**

Theorem 6.3 (Eulerian graphs of maximum degree 6)**.** *Let G be Eulerian with $\Delta(G) \leq 6$. Then: (i) $\mathrm{ce}(G) \leq 3\lfloor n/3 \rfloor \leq n$; (ii) if $n \not\equiv 0 \pmod 3$, or if G is not 6-regular, then $\mathrm{ce}(G) \leq n - 1$. In particular, $\mathrm{ce}(G) \geq n$ forces $n \equiv 0 \pmod 3$ and G 6-regular.*

Proof sketch. By [1] (Thm. 1.5: an even graph of maximum degree Δ decomposes into $\Delta/2$ 2-regular subgraphs), $E(G)$ splits into F_1, F_2, F_3. Each F_i is a union of vertex-disjoint cycles of length at least 3, so F_i contributes at most $\lfloor n/3 \rfloor$ parts, and the cycles of the three factors concatenate to a decomposition. If $n = 3m' \pm 1$ this already gives $\leq n - 1$. If G is not

6-regular, some vertex has degree at most 4 and is missed by one of the factors, saving one further part. □

Proof status: Complete proof in campaign records, `campaign/lane_05_delta6.md` Thm. A..

So everything funnels into one case: 6-regular, $n \equiv 0 \pmod 3$. Theorem 6.3 leaves it at ce $\leq n$; we now close it at $n-2$. Throughout, 6-regular graphs have $n \geq 7$, so under $n \equiv 0 \pmod 3$ we have $n \geq 9$; write $n = 3m$.

Lemma 6.4 (Petersen's 2-factorisation, self-contained form)**.** *Every $2k$-regular graph decomposes into k edge-disjoint spanning 2-factors.*

Proof sketch. Orient each component along an Euler tour; every vertex gets in-degree = out-degree = k. Split each vertex into an out-copy and an in-copy and view the arcs as edges of a k-regular bipartite graph; by Hall's theorem this graph has a perfect matching, and removing perfect matchings one at a time gives k edge-disjoint perfect matchings (König). Each matching is a 1-in/1-out spanning digraph without 2-cycles (each edge is oriented once), hence a union of directed cycles of length at least 3; its underlying undirected graph is a spanning 2-factor. □

Proof status: Complete proof in campaign records, `campaign/r2_01_aabc25_struct.md` §1, Lemma 0; classical, included for completeness..

Lemma 6.5 (Rigidity of triangle-forced 4-regular graphs)**.** *Let G be a 4-regular simple graph with $|V(G)| \geq 5$ in which every 2-factor is a union of triangles. Then there are a cubic bipartite graph B and an isomorphism $G \cong L(B)$. In particular G has two edge-disjoint triangle 2-factors, and any two of its triangles from different factors share at most one vertex.*

Proof sketch. Suppose every 2-factor of G is a union of triangles, and argue in steps. (1) By Lemma 6.4 with $k = 2$, $G = F \sqcup F'$ with F a 2-factor; then $F' = G - E(F)$ is also a 2-factor, and both are triangle 2-factors. Write $n = 3m$, $m \geq 2$. (2) Crossing pattern. If $a \neq b$ lay in both a triangle τ of F and a triangle σ of F', then, triangles being cliques in a simple graph, the edge ab would lie in $E(F) \cap E(F')$, which is empty; so $|V(\tau) \cap V(\sigma)| \leq 1$. Each vertex lies in exactly one triangle of each factor. If two vertices of τ sat in the same F'-triangle, that triangle would share two vertices with τ; hence the three vertices of τ lie in three *distinct* F'-triangles, and symmetrically for σ. (When $m = 2$ this is already a contradiction, so $m \geq 3$.) (3) Incidence graph. Build a bipartite graph B with sides the triangles of F and of F', and one edge of B for each vertex v of G, joining $\tau(v)$ to $\sigma(v)$. Step (2) says B is simple and cubic; it is bipartite by construction. Edges of G are exactly the pairs of vertices sharing a triangle, i.e. pairs of B-edges sharing a B-endpoint, so $G \cong L(B)$. □

Proof status: Complete proof in campaign records, `campaign/r2_01_aabc25_struct.md` §2, Lemma 1, steps (1)–(3); full proof in campaign records. The structural reduction appears to have no precedent; the exhaustive verification below covers every potential counterexample precisely because of it..

Proposition 6.6 (The two complementary 2-factors of a line graph)**.** *Let B be a cubic bipartite simple graph, let $W \cong L(B)$, let M be a perfect matching of B, and let σ be an orientation of M, with $-\sigma$ the reversed orientation. Then $E(\Phi(\sigma))$ and $E(\Phi(-\sigma))$ partition $E(W)$, and both $\Phi(\sigma)$ and $\Phi(-\sigma)$ are triangle-free 2-factors of W (all cycles of length ≥ 4).*

Proof sketch. For each $z \in V(B)$ let $\{e_z, f_z, g_z\}$ be the three incident edges, e_z being the matched one, and let P_z be the three pairs of edges sharing z. Since B is simple, each edge of $W = L(B)$ is a pair of B-edges sharing a *unique* endpoint, so the P_z partition $E(W)$. Now read pointwise: at a tail of σ, $\Phi(\sigma)$ takes the pairs $\{e_z f_z, e_z g_z\}$; at a head, it takes $\{f_z g_z\}$; and $\Phi(-\sigma)$ does the opposite. In both cases the two factors take distinct pairs summing to three per P_z, so their edge sets partition $E(W)$. A degree check at each B-edge — a

matched edge gets $2+0$ neighbours, an unmatched edge $1+1$ — shows both are 2-regular and spanning. For triangle-freeness: a triangle of $L(B)$ corresponds to three B-edges meeting at one point, and at that point one of the three pairs is precisely the pair not taken by the factor in question; B has no triangles of its own, being bipartite. □

Proof status: Complete proof in campaign records, `campaign/r2_01_aabc25_struct.md` §2, Proposition 6 and steps (4)–(8); full proof in campaign records. Positioning: the existence statement — a cubic graph has a perfect matching iff its line graph admits a triangle-free 2-factorisation — is an observation of Kouider–Sabidussi [20], and the case proved here (cubic bipartite B) is a direct special case of it via Hall's theorem; we claim no priority for it. The explicit pointwise partition $\Phi(\sigma)\sqcup\Phi(-\sigma)$, its "for every orientation" quantifier, and the counting corollary below are the contribution claimed here. See also Bertram–Horák [4] for a polynomial decision algorithm for the general 4-regular case..

Theorem 6.7 (Six-regular graphs, strengthened bound)**.** *Every 6-regular simple graph G with $|V(G)| = n = 3m \geq 9$ satisfies $\mathrm{ce}(G) \leq n-2$.*

Proof sketch. Take any 2-factorisation $G = F \sqcup F' \sqcup F''$ (Lemma 6.4, $k=3$). Case 1: G has no triangle 2-factor. Then none of the three factors is a union of triangles, so each contains a cycle of length ≥ 4 and has at most $1 + \lfloor (n-4)/3 \rfloor = m-1$ components; the total is at most $3m-3 = n-3$. Case 2: G has a triangle 2-factor S, with exactly m parts, and $H := G - E(S)$ is a 4-regular spanning simple graph on $n \geq 9$ vertices. — If H has a triangle 2-factor T, then S and T are edge-disjoint triangle 2-factors of a 4-regular spanning subgraph $W = S \sqcup T$, and the rigidity lemma gives $W \cong L(B)$; Proposition 6.6 splits W into two triangle-free 2-factors with at most $m-1$ parts each, while $R := G - E(W)$ is a 2-factor with at most m parts. Total: $(m-1)+(m-1)+m = 3m-2 = n-2$. — If H has no triangle 2-factor, take any 2-factorisation $H = T \sqcup U$: neither is a triangle union, so each has at most $m-1$ parts, and the total is $m+(m-1)+(m-1) = n-2$. □

Proof status: Complete proof in campaign records, `campaign/r2_01_aabc25_struct.md` §3, Theorem 2′; full proof in campaign records. Numerics (Appendix A): exhaustive verification of the structural lemma's full parameter space for $s \leq 5$ (832,944 combinations) and the $L(B)$ family $s = 3..6$ (299,265 instances); all 30,016 labelled 6-regular graphs on 9 vertices satisfy the bound with room to spare — each has exact $\mathrm{ce} = 3$..

Theorem 6.8 (Conjecture 6.2 of [1])**.** *Every 6-regular simple graph decomposes into three edge-disjoint spanning 2-factors, one of which has a component with at least 4 vertices.*

Proof sketch. Three cases. If $3 \nmid n$, no triangle 2-factor exists (it would force $3 \mid n$), so any 2-factorisation from Lemma 6.4 has a non-triangle factor, which contains a cycle of length ≥ 4. If $3 \mid n$ and G has no triangle 2-factor, the same applies to every factor of any 2-factorisation. If $3 \mid n$ and G has a triangle 2-factor S, then $|V(G)| = n \geq 9$ and $H := G - E(S)$ is 4-regular on at least 5 vertices; by the contrapositive of Lemma 6.5, H has a 2-factor S' containing a cycle of length ≥ 4, and $S \sqcup S' \sqcup T$ with $T := H - E(S')$ is the required decomposition. □

Proof status: Complete proof in campaign records, `campaign/r2_01_aabc25_struct.md` §3, Theorem 3 (the three-case decision written out in full); full proof in campaign records..

Corollary 6.9 (Six-regular graphs)**.** *Every 6-regular simple graph G on n vertices satisfies $\mathrm{ce}(G) \leq n-1$. If $n \equiv 0 \pmod 3$ this holds in the stronger form $\mathrm{ce} \leq n-2$ (Theorem 6.7); if $n \equiv 2 \pmod 3$ it holds in the form $\mathrm{ce} \leq n-2$ as well, via $3\lfloor n/3 \rfloor = n-2$.*

Corollary 6.10 (Even graphs of maximum degree six)**.** *Every even graph with $\Delta \leq 6$ satisfies $\mathrm{ce}(G) \leq n-1$: apply Theorem 6.3(ii) away from the exceptional case and Corollary 6.9 in it. Together with Theorem 6.1 this locates the failure of the $n-1$ budget at $\Delta \leq 6$ exactly in the presence of odd degrees.*

Remark 6.11 (Tightness is open, and the truth is far lower). We do not know whether $n-1$ is attained in the even world: the path P_n has $\mathrm{ce}(P_n) = n-1$ but is not an even graph, so it is no witness here. The computational record suggests the truth is far below the proof: all 6-regular graphs on 9, 10 vertices have exact ce = 3, and $n = 12$ has 7,849 isomorphism classes whose decomposition ledgers reach $n-2 = 10$ pieces in only 5 classes — and even those have exact ce = 3. The proof closes at $n-2$ through the ledger, not through the truth. At degree 8 the analogous bound is immediate from the all-2-factors-isomorphic analysis of [2], as noted in [1]; degree 6 is the last even degree where a structural argument of the kind given here is needed. Whether the two residual configurations of Theorem 6.7's case analysis can be pushed to $n-3$ is registered as an open problem in Section 11.

6.3. The general $\Delta \le 6$ question.

Theorem 6.12 (General $\Delta \le 6$ graphs)**.** *If $\Delta(G) \le 6$ and $\tau = \tau(G)$, then $\mathrm{ce}(G) \le \tau + \lfloor (m-\tau)/3 \rfloor$ with $m \le 3n$. Consequently: $\tau \le 3n/4 \Rightarrow \mathrm{ce} \le 3n/2$; $\tau \le n/2 \Rightarrow \mathrm{ce} \le 7n/6$; and always $\mathrm{ce} \le 5n/3 - 2/3$.*

Proof status: Complete proof in campaign records, `campaign/lane_05_delta6.md` Thm. D, via Lemma 2.5 with $m \le 3n$..

Remark 6.13 (An honest gap). We do not claim $\mathrm{ce} \le n$ for all $\Delta \le 6$. The even case is now closed (Corollary 6.10), but when the graph has odd degrees and $\tau \in (3n/4, n-1]$ with $m > 3n - 2\tau$ — long T-joins on dense graphs — no proof of $\mathrm{ce} \le n$ is known to us; this mirrors the "tree-absorbing rewind" obstruction identified for $\Delta = 5$ in the records. The counterexample of Theorem 6.1 sits exactly in such a regime: the extremal accounting cannot close, yet the true value is $\mathrm{ce} = n$.

Remark 6.14 (The $K_{3,n-3}$ mechanism). For $n \ge 7$, $\mathrm{ce}(K_{3,n-3}) \ge n-1$: the $n-3$ vertices of degree 3 force $n-3$ single edges, and the remaining $2(n-3)$ edges exceed the capacity n of a single cycle, forcing at least two cycle parts; the exact values $\mathrm{ce}(K_{3,4}) = 6$, $\mathrm{ce}(K_{3,5}) = 7$, $\mathrm{ce}(K_{3,6}) = 8$ all equal $n-1$. The case $n = 6$ is a genuine exception: $\mathrm{ce}(K_{3,3}) = 4 = n-2$, because the six residual edges fit in a single Hamilton cycle. This parity mechanism, "upgraded" by the triangle of Theorem 6.1, is the principal source of $\Delta \le 6$ lower bounds.

7. Odd degrees: the cubic case of Conjecture 6.4

The paper [1] also studies a relaxation. An $\mathrm{f_{re}}$-*decomposition* of G is a partition of $E(G)$ into blocks, each block being the edge set of a 2-regular subgraph (a union of vertex-disjoint cycles) or a single edge, and $\mathrm{f_{re}}(G)$ is the minimum number of blocks. Every ce-decomposition is an $\mathrm{f_{re}}$-decomposition, so $\mathrm{f_{re}}(G) \le \mathrm{ce}(G)$; what the relaxation buys is that several cycles may share one block. Conjecture 6.3 of [1] proposes $\mathrm{f_{re}}(G) = O(n)$ for all graphs, and Conjecture 6.4 there proposes $\mathrm{f_{re}}(G) \le n-1$ for every $(2k+1)$-regular graph. The even-regular case is settled there (Thm. 1.5: $\mathrm{f_{re}}(G) = \Delta/2$ for an even graph of maximum degree Δ), and the same paper observes that a perfect matching settles the odd-regular case: taking the matching as single edges, $\mathrm{f_{re}} \le n/2 + k$. The open core is thus the odd-regular graphs *without* a perfect matching. This section settles the cubic case completely.

Theorem 7.1 (Reduction identity)**.** *For every graph G with m edges,*

$$\mathrm{f_{re}}(G) \;=\; \min_{H} \big(m - |E(H)| + \Delta(H)/2\big),$$

where the minimum is over even subgraphs $H \subseteq G$.

Proof sketch. ($\le$) Given H, the construction behind [1] Thm. 1.5 produces $\Delta(H)/2$ spanning subgraphs of H, each of every-vertex-degree 0 or 2; together with the $m - |E(H)|$ edges of $G - H$ as single edges this is an $\mathrm{f_{re}}$-decomposition with the displayed count. ($\ge$) From an optimal $\mathrm{f_{re}}$-decomposition with t two-regular blocks and s single edges, let H be the union of the blocks. A vertex lying in r_v blocks has $\deg_H(v) = 2r_v$, so H is even and $\Delta(H)/2 = \max_v r_v \le t$; hence $s + \Delta(H)/2 \le s + t = \mathrm{f_{re}}(G)$. □

Proof status: Complete proof in campaign records, `front_tre/attack_tre.md` §3, Theorem A; full proof in campaign records..

Let $p(G)$ denote the minimum size of a parity subgraph with $T = V(G)$, and let $p_2(G)$ denote the maximum size of an even subgraph. Two immediate consequences of Theorem 7.1: if G is $(2k+1)$-regular then $\mathrm{f_{re}}(G) \le p(G) + k$ (remove a minimum parity subgraph); and on cubic graphs the formula sharpens to an equality, since $G - F$ has degrees in $\{0, 2\}$:

Theorem 7.2 (Cubic exact formula)**.** *If G is cubic, then $\mathrm{f_{re}}(G) = p(G) + 1$, and $p(G) = m - p_2(G)$; consequently Conjecture 6.4 of [1] for cubic G is equivalent to $p_2(G) \ge n/2 + 2$.*

Proof sketch. For a minimum parity subgraph F, the complement $G - F$ is an even subgraph of a subcubic graph, hence a disjoint union of cycles — a single two-regular block when nonempty — giving $\mathrm{f_{re}} \le p + 1$. Conversely G contains a cycle C, and $G - E(C)$ is an odd factor of size $m - |C| \le m - 3$, so $p \le m - 3 < m$ and the minimum in Theorem 7.1 is never attained with $H = \emptyset$; the exact value is $p + 1$. For $p = m - p_2$: the complement of a maximum even subgraph is a minimum T-join and vice versa. □

Proof status: Complete proof in campaign records, `front_tre/attack_tre.md` §3, Corollary A3; full proof in campaign records..

The perfect-matching case of the cubic conjecture is now immediate ($\mathrm{f_{re}} \le n/2 + 1 \le n - 1$ for $n \ge 4$), and this is the observation already recorded in [1]. The substance is the graphs without a perfect matching. Here the Petersen theorem in its contrapositive form is the structural entry point: a cubic graph without a perfect matching has a bridge. Bridges do not occur inside blocks, and T-joins decompose over the block tree — which is what the next two results exploit.

Lemma 7.3 (Bridge decomposition)**.** *Let G be connected cubic with bridge set Br, $B = |Br|$, and let $X_1, \dots, X_k$ be the components of $G - Br$, each bridgeless and subcubic; write b_v for the number of bridges at v and let $T_i = \{v \in X_i : b_v \text{ even}\}$. Then $|T_i|$ is even, and*

$$p(G) \;=\; \sum_i q(X_i) \;+\; B, \qquad q(X_i) := \min\{|F| : F \text{ a } T_i\text{-join in } X_i\}.$$

Proof sketch. Every bridge lies in every odd spanning factor: if a bridge e cuts off A, then $|A|$ is odd (since $3|A| - 1 = 2e(A)$), and $F \cap E(A)$ would have all $|A|$ vertices odd — impossible by the handshake lemma. So B edges are forced. Within X_i, the requirement $\deg_F \equiv 1 \pmod 2$ becomes $\deg_{F \cap X_i} \equiv 1 + b_v \pmod 2$, i.e. $F \cap X_i$ is a T_i-join; and $|T_i|$ is even because $3n_i - \sum_v b_v = 2e(X_i)$ forces $\sum b_v \equiv n_i \pmod 2$. Both directions are now immediate, and the minima add. □

Proof status: Complete proof in campaign records, `front_tre/attack_tre.md` §4, Theorem C; full proof in campaign records..

Lemma 7.4 (Piece lemma; Choi–Kim–Kostochka–Park–West)**.** *Let X be connected, bridgeless and subcubic with $n \ge 3$ vertices. Then $p_2(X) \ge n/2 + 3/2$. More generally, in the setting of Lemma 7.3, writing $\Phi(X) := n - q(X)$ and b for the number of bridge-ends at vertices of X, one has $\Phi(X) \ge b/2 + 3/2$.*

Proof sketch. On subcubic graphs, maximum even subgraphs are exactly maximum 2-regular subgraphs counted by edges, so p_2 equals the quantity f_2 of [9], whose Theorem 1.3 gives $f_2 \ge n - \max\{0, (d + c - 1)/2\}$ in terms of the number c of cut-edges and the 1-deficit $d = 3n - 2m$. Bridgelessness gives $c = 0$, and the three branches are: all degrees 2 (a cycle: $p_2 = n$); all degrees 3 ($d = 0$: $p_2 = n$, which is Petersen's 1-factor theorem); mixed ($d \ge 1$ and $m \ge n + 1$, giving $p_2 \ge n - (d-1)/2 = m - n/2 + 1/2 \ge n/2 + 3/2$). For the corollary form, $X - H$ is a T-join for H maximum even, so $q \le m - p_2$ and $\Phi = n - q \ge n - m + p_2 \ge (3n - 2m)/2 + 3/2 = b/2 + 3/2$, using $\sum_v \deg_X = 3n - b = 2m$. □

Proof status: This is a quoted published theorem plus a conversion derived in campaign records, `front_tre/attack_tre.md` §4 (Lemma A = [9] Thm. 1.3, whose own proof uses Edmonds' weighted matching and the theorem of Plesník; Lemma B = the conversion; full details in campaign records). The tight example is $K_{2,3}$: $p_2 = 4 = n/2 + 3/2$..

Lemma 7.5 (Assembly)**.** *Let G be connected cubic without a perfect matching, with block tree having k blocks ($B = k - 1$ bridges): r non-trivial blocks on $n \geq 4$ vertices, c triangle blocks, and t single-vertex blocks. Then $r \geq c + t + 2$, and $\sum_X \Phi(X) \geq B + 2$.*

Proof sketch. Degree conservation over the block tree gives $\sum_X b(X) = 2B = 2(k-1)$, where $k = r + c + t$ is the number of blocks; every block has $b \geq 1$, and triangle and single-vertex blocks have $b = 3$. Hence $2(r + c + t - 1) \geq 3(c + t) + r$, i.e. $r \geq c + t + 2$. For the ledger: non-trivial blocks give $\Phi \geq b/2 + 3/2$ by Lemma 7.4, triangle blocks give $\Phi = 3 = b/2 + 3/2$, single-vertex blocks give $\Phi = 1$; summing,

$$\textstyle\sum_X \Phi \;\geq\; \sum_{\text{pos}}(b_i/2 + 3/2) + 3c + t \;=\; B + 1.5r + 1.5c - 0.5t \;\geq\; B + 2,$$

the last step by $r \geq c + t + 2$. □

Proof status: Complete proof in campaign records, `front_tre/attack_tre.md` §4 (assembly lemma); full proof in campaign records..

Theorem 7.6 (Cubic graphs)**.** *Every connected cubic graph without a perfect matching satisfies $p(G) \leq n - 2$, hence $\mathrm{f_{re}}(G) = p(G) + 1 \leq n - 1$. Together with the perfect-matching case, every cubic graph G on $n \geq 4$ vertices satisfies $\mathrm{f_{re}}(G) \leq n - 1$: Conjecture 6.4 of [1] holds for $k = 1$.*

Proof sketch. Parts never straddle components, so $\mathrm{f_{re}}$ is additive over components and we may assume G connected. If G has a perfect matching, remove it: $\mathrm{f_{re}} \leq n/2 + 1 \leq n - 1$ for $n \geq 4$. Otherwise, by Petersen's theorem in the contrapositive, G has a bridge, so $k \geq 2$. Lemma 7.3 writes $p = \sum_i q_i + B$, and $p = \sum_i q_i + B = n - \sum_X \Phi(X) + B$, so the assembly lemma gives $p \leq n - (B + 2) + B = n - 2$. □

Proof status: Complete proof in campaign records, `front_tre/attack_tre.md` §4, Theorem D; full proof in campaign records. Honest scope: the proof relies on the published Theorem 1.3 of [9]; the contribution claimed here is the reduction chain (Theorem 7.1, Lemma 7.3, Lemma 7.5), the exhaustive verification below, and the structural analysis of the small counterexamples. Computational verification: all 9,588 connected odd-regular graphs on $n \leq 12$ vertices (degrees 3 through 11), and all connected cubic graphs on $n = 14, 16, 18$ vertices (509, 4,060, 41,301 graphs), with zero violations; all 60,077 bridgeless subcubic graphs on $n \leq 14$ vertices satisfy Lemma 7.4 with a minimum margin of exactly 3/2, attained uniquely by $K_{2,3}$ (Appendix A)..

Remark 7.7 (The smallest imperfect cubic graphs). The minimum order of a cubic graph without a perfect matching is 16, and the extremal graph is unique; the minimality is classical — it follows from work of Errera and of Chartrand–Goldsmith–Schuster [7] — and our contribution is the enumeration confirming uniqueness up to isomorphism for all orders ≤ 18, plus the structure: the extremal graph is three subdivided-K_4 blades joined to one central vertex by three bridges, and it has $p = n/2 + 1$. In fact every perfect-matching-free cubic graph found in the exhaustive sweeps has $p = n/2 + 1$, with the excess over $n/2$ coming entirely from single-vertex blocks. For $k \geq 2$ the whole question reopens: without regularity 3, "no perfect matching" no longer forces a bridge, and the correct cut structure is that of Tutte sets; a Tutte-obstruction family of 5-regular graphs (orders 36 to 146) satisfies $p \leq n - 3$ with room to spare, and the general case is open.

8. Bounded degeneracy

Theorem 8.1 (Degeneracy 2: the exact frontier)**.** *Let G be 2-degenerate with $c = c(G)$ components and $s(G)$ non-tree components. Then $\mathrm{ce}(G) \le n - c - s(G) \le n - 1$, with $\mathrm{ce}(G) = n - 1$ if and only if G is a tree. In particular every K_4-minor-free (equivalently, treewidth-≤ 2) graph satisfies $\mathrm{ce}(G) \le n - 1$.*

Proof sketch. Induction on connected G via a vertex v with $\deg(v) \le 2$. A bridge at v is charged exactly: it becomes a forced single edge. If the two v-edges' other endpoints u, w are adjacent, both edges are absorbed into the part containing uw at no cost; if not, dissolve v into the chord uw (solvability preserves 2-degeneracy), paying at most one. Components add by Corollary 3.4's ledger. Non-tree components pay their extra 1; hence equality in $\mathrm{ce} \le n - 1$ occurs exactly for connected forests, i.e. trees. □

Proof status: Complete proof in campaign records, `campaign/lane_08_degeneracy.md` Thm. 2.1 with Lemma 2.3 (dissolution lemma); 725 randomized 2-degenerate instances machine-checked (Appendix A)..

Theorem 8.2 (Degeneracy 3: an upper bound)**.** *If G is 3-degenerate and $n \ge 3$, then $\mathrm{ce}(G) \le \lfloor (5n - 8)/3 \rfloor$; connectedness may be replaced by the sharper form with $n - c(G)$. More generally, d-degenerate graphs satisfy $\mathrm{ce}(G) \le \lfloor ((d+2)n - d(d+1)/2 - 2)/3 \rfloor$ and $\mathrm{ce}(G) \le m$.*

Proof sketch. A degeneracy ordering gives $m \le dn - d(d+1)/2$; combine with Lemma 2.5 and $\tau \le n - c$. □

Proof status: Complete proof in campaign records, `campaign/lane_08_degeneracy.md` Thm. 3.3 (T3a), using the independently re-proven Lemma 2.5..

Theorem 8.3 (Exact values for $K_{3,q}$)**.** *For every $q \ge 2$,*

$$\mathrm{ce}(K_{3,q}) = \left\lceil \tfrac{4q}{3} \right\rceil.$$

Consequently $\sup\{\mathrm{ce}(G)/n : G \text{ 3-degenerate}\} \ge 4/3$, attained along $q \equiv 0 \pmod 3$.

Proof sketch. Upper bound: split $K_{3,q}$ into blocks $K_{3,3}$ (C_6 plus 3 single edges, 4 parts), handling $q = 3k, 3k+1, 3k+2$ with tail blocks $K_{3,4}$ (6 parts) and $K_{3,2} = C_4$ plus 2 single edges (3 parts); each case sums to $\lceil 4q/3 \rceil$. Lower bound (slot–rate lemma): every cycle C in $K_{3,q}$ has $|C| - 1 \le \frac{5}{3}|C \cap B|$, and each degree-3 vertex of the large side feeds at most one cycle; hence $\mathrm{ce} \ge 3q - \lfloor \frac{5}{3} q \rfloor = \lceil 4q/3 \rceil$. □

Proof status: Complete proof in campaign records, `campaign/lane_08_degeneracy.md` Thm. 3.5 (T3c); an arithmetic slip in the $q \equiv 2$ line of the first draft was corrected in review ($K_{3,2}$ contributes 3 parts, not 1), the total being unchanged..

Conjecture 8.4 (Matching upper bound for $d = 3$)**.** *Every 3-degenerate graph satisfies $\mathrm{ce}(G) \le \lfloor (4n - 6)/3 \rfloor$.*

Remark 8.5. Conjecture 8.4 is equivalent, via the peeling ledger, to a $\tau \le (4n - 6 - m)/2$-type bound on 3-cores; 630 randomized 3-degenerate instances with exact ce and exact τ showed zero violations. The state of knowledge here deserves emphasis: the lower bound 4/3 for $\sup \mathrm{ce}/n$ is proved, with an exact extremal family (Theorem 8.3); the matching upper bound is open, and the best proved upper bound is 5/3 (Theorem 8.2).

Theorem 8.6 (Failure at every $d \ge 3$)**.** *For every $d \ge 3$ there are d-degenerate graphs, on unboundedly many vertices, with $\mathrm{ce}(G) > n - 1$: (i) $d = 3$: $\mathrm{ce}(K_{3,7}) = 10 = |V(K_{3,7})|$; (ii) $d = 3$, non-bipartite: for $q \ge 14$, $G_q = K_{3,q} + xy$ (one edge inside the large side) satisfies $\deg(G_q) = 3$ and $\mathrm{ce}(G_q) \ge (4q - 7)/3 > |V(G_q)| - 1$; (iii) $d = 5$: $\mathrm{ce}(K_{5,11}) \ge 16 > 15$; (iv) in general, for odd $d = 2k + 1$ and q large, $\mathrm{ce}(K_{2k+1,q}) \ge \frac{3k+1}{2k+1} q - O(1) > n - 1$.*

Proof sketch. All items use the slot–rate lemma: if every cycle C satisfies $|C|-1 \le \rho|C\cap B|$, then any decomposition has at least $m - \rho\sum_{v\in B}\lfloor \deg(v)/2\rfloor$ parts, since each $v \in B$ lies in at most $\lfloor \deg(v)/2\rfloor$ cycles. In $K_{3,q}$ every cycle has $|C| - 1 \le \frac{5}{3}|C \cap B|$ and the large side supplies exactly q slots; adding xy relaxes the rate inequality only within the same constant, and $\deg(x) = \deg(y) = 4$ adds two slots, whence (ii). For (iv) the constant $(3k+1)/(2k+1) \to 3/2$ matches the global Erdős lower bound, so these families are extremal simultaneously for their degeneracy level and for the whole problem. □

Proof status: Complete proof in campaign records, `campaign/lane_08_degeneracy.md` *Thms. 4.2–4.3, with machine certificates: for* $K_{3,14}+e$ *all* 3,324 *cycles were enumerated for the rate inequality (*$\mathrm{ce} \in [17, 21]$*), for* $K_{5,11}$ *all* 794,530 *cycles (*$\mathrm{ce} \ge 16$*);* $\mathrm{ce}(K_{3,7}) = 10$ *by exact solver (Appendix A)..*

Remark 8.7 (The degeneracy frontier). Within the degeneracy hierarchy the picture is complete: $n - 1$ holds for all 2-degenerate graphs (Theorem 8.1) and fails by arbitrarily large margins at every $d \ge 3$ (Theorem 8.6). This kills any goal of the shape "degeneracy ≤ 6 $\Rightarrow n - 1$ up to finitely many graphs". Note the complementarity with Section 6: the counterexample there lives in the Δ-bounded world, Theorem 8.6 in the degeneracy-bounded, Δ-unbounded one; the common enemy is the odd complete-bipartite kernel.

9. Extremal examples and the shape of $f(n)$

Theorem 9.1 (Complete bipartite lower bounds; restated)**.** *Let $1 \le s \le t$. Then: (i) if s is odd,* $\mathrm{ce}(K_{s,t}) \ge t(3s-1)/(2s)$*; (ii) if s is even and t is odd,* $\mathrm{ce}(K_{s,t}) \ge (2s+t-1)/2$*; (iii) if s, t are even,* $\mathrm{ce}(K_{s,t}) \ge t/2$*. In particular $f(n) \ge (3/2 - o(1))n$. No novelty is claimed: case (i) and the $(3/2 - o(1))n$ consequence are known [13, 6], and are re-derived here by a degree-budget argument; cases (ii)–(iii) are footnote-level refinements recorded in the archive.*

Proof status: Complete proof in campaign records, `notes/attack_bipartite.md` *Thm. 1.1 (with small-value table:* $\mathrm{ce}(K_{2,5}) = 4$*,* $\mathrm{ce}(K_{3,4}) = 6$*,* $\mathrm{ce}(K_{3,7}) = 10$*,* $\mathrm{ce}(K_{5,5}) = 7$*, exact)..*

Theorem 9.2 (A non-bipartite exact family)**.** *Let $G_c = K_2 \vee K_{1,c-1}$ (two universal vertices joined to a star with $c-1$ leaves), $c \ge 3$, $n = c+2$. Then*

$$\mathrm{ce}(G_c) = c - 1 + \left\lceil \tfrac{c+2}{3} \right\rceil = n - 3 + \lceil n/3 \rceil \approx \tfrac{4}{3}n.$$

Proof status: Complete proof in campaign records, `campaign/lane_03_constant.md` *Thm. M2 (two-sided accounting on the savings identity; exact for all c, cross-checked at $c = 2, \ldots, 8$ with multiple realizations and dual certificates)..*

Corollary 9.3. *For every $n \ge 7$ there is an n-vertex graph with* $\mathrm{ce}(G) \ge n$*; i.e. $f(n) \ge n$ for all $n \ge 7$ (with $f(10) \ge 11$, $f(11) \ge 12$). Two independent routes are known: Theorem 9.2 and the $K_{3,q}$ route (Theorem 8.3 for $n \ge 10$).*

Proof status: Complete proof in campaign records, `campaign/lane_03_constant.md` *Cor. N.1..*

Proposition 9.4 (Small 4-regular graphs)**.** *Every 2-connected 4-regular graph on $n \le 8$ vertices satisfies* $\mathrm{ce}(G) \le n/2 - 1$*.*

Proof status: Complete proof in campaign records, `campaign/lane_04_delta5_savings.md` *C2: $n = 6$ has a unique isomorphism class; $n = 8$ follows from Dirac's theorem ($2\delta = 8 \ge n$, so the graph is Hamiltonian, giving $1 + \lfloor 8/3 \rfloor = 3$). The case $n \ge 10$ is honestly open: it would require a cycle of length $\ge n/2 + 6$ in the residual..*

Remark 9.5 (Where does $n - 1$ bite?). Three structurally different mechanisms push ce to its budget: bridges (trees); parity at density (even complete graphs, Theorem 3.1); and concentrated odd degree in near-bipartite kernels ($K_{3,4}$, $K_{3,q}$: Theorems 8.3 and 6.1). A structural characterisation of the graphs with $\mathrm{ce}(G) = n - 1$ is a central open problem (Section 11); the tight-instance census for $\Delta \le 5$, $n \le 7$ (seven 2-connected isomorphism

classes) shows that all small tight instances fall into the third mechanism or are small dense graphs.

10. Anatomy of the $\log^*$ barrier

We recall the skeleton of [6] in the form needed below. A graph G is an (ε, s)-*expander* if $|N_{G-F}(U)| \geq \varepsilon|U|/\log^2 n$ for every U with $|U| \leq 2n/3$ and every F with $|F| \leq s$. BM splits the vertex set into parts that are (ε, s)-expanders, at a cost of $4sn \log n$ deleted edges with $s = \log^{273} n$, then extracts cycles from each expander, the remainder after each extraction round having average degree at most $f(d) = O(\log^{274} d)$; iterating against a fresh tower reference ("tower reset") gives $O(\log^* n)$ rounds and hence $O(n \log^* n)$ total. All statements below are *within this architecture*: they concern what any refinement of the parameters, or of the single-step density lemma, can achieve. They are not unconditional lower bounds against every conceivable algorithm.

Proposition 10.1 (Fixed-point lock-in)**.** *Let $f\colon [d_0, \infty) \to [1, \infty)$ be nondecreasing.*

(a) *If f has a (necessarily superconstant) fixed point $d^* > d_0$ with $f(d^*) \geq d^*$ — as happens for every $f(d) = (\log d)^c$, $c \geq 1$; for $c = 274$ the fixed point of $x = 274 \ln x$ is $x^* \approx 2095.4$ — then naive iteration $f^i(n)$ never drops below d^*: the single-step lemma alone cannot terminate.*

(b) *(Tower-step condition.) Suppose there is a monotone g and a constant $C > 0$ with $f(Cg(t)) \leq Cg(t/2)$ for all large t. Then any decomposition process whose only per-step information is "remainder average degree $\leq f(\text{current})$" and which may reset against the original n has depth $\leq \log^* n + O(1)$. The condition is satisfied by polylogarithmic f (this is exactly BM's bookkeeping), but not in general: for $f(d) = d/2$ the depth is $\Theta(\log n)$, and no reset changes that.*

Proof sketch. (a) Monotone iteration converges down to the unique fixed point. (b) Induction: $d_i \leq Cg(\log^{[i]} n)$, each step lowering the tower by one level; this abstracts the induction in [6], verified against the source. □

Proof status: Complete proof in campaign records, `campaign/lane_12_expansion.md` Prop. 12.1. Part (b)'s statement is the reviewer-narrowed form: for general f the claim is false, and the tower-step hypothesis is load-bearing..

Proposition 10.2 (The phase transition is a mirage)**.** *Let $\tilde{f}(d) = 2^{(\ln d)^\theta}$, $0 < \theta < 1$. Then the naive iteration depth of $\tilde{f}$ from n is $\Theta(\ln\ln\ln n/\ln(1/\theta))$, which exceeds $\log^* n$ for every n with $\log^* n \geq 4$; and $\log^* n = o(\log\log\log n)$. Consequently, a single-step lemma with remainder $2^{(\log d)^\theta}$ would yield $O(n \log\log\log n)$ — a bound weaker than the existing $O(n \log^* n)$. The "$2^{(\log d)^\theta}$ rung" sits between the CFS rung ($d^{1-\delta}$, depth $\Theta(\log\log n)$) and the BM rung (polylog, depth $\log^* n$): it is the next rung down, not up.*

Proof sketch. Set $x_i = \ln\ln \tilde{f}^{\,i}(n)$; then $x_{i+1} = \theta x_i + o(1)$. For the comparison, unwind the tower: $\log^* n = k$ forces $n > 2 \uparrow\uparrow (k-1)$, so $\ln\ln\ln n \gtrsim 2 \uparrow\uparrow (k-4)$. Numerically at $k = 4$: $\ln\ln n \approx 10.72$, $\ln\ln\ln n \approx 2.37$, depth $\approx 22.5 > 4$. The moral: iteration depth is a dynamical quantity, and must not be inferred from the pointwise size of the remainder map. □

Proof status: Complete proof in campaign records, `campaign/lane_12_expansion.md` Prop. 12.2 (constants corrected in review: crossover $\ln d \approx 5607$ for $\theta = 0.9$; fixed point $x^* \approx 2095.4$, not $274 \ln 274$). An earlier, opposite interpretation of an archive phase-transition claim was withdrawn; see Acknowledgments..

Proposition 10.3 (Self-improvement is impossible by deletion alone)**.** (a) *If $G - F$ is an (ε, s)-expander for some $F \subseteq E(G)$, then $\delta(G) \geq s + 1$. Hence no graph of minimum degree $\leq s$ can be turned into an (ε, s)-expander by deleting edges.*

(b) *(Finite witness, downgraded form.) The path P_4 is an $(\varepsilon_0, 0)$-expander for every $0 < \varepsilon_0 \le 24/25$ (natural logarithms) and has $\delta(P_4) = 1$; this is verified by exhaustive check of all admissible vertex subsets.*

(c) *Combining (a) and (b): there exist minimum-degree-1 $(\varepsilon_0, 0)$-expanders that admit no edge-deleted (ε', s)-expander for any $s \ge 1$. In BM's architecture, vertex splitting plus edge deletion is a necessity, not a technical choice.*

Proof sketch. (a) Delete, at a minimum-degree vertex v of $G - F$, all its incident edges; expansion applied to $U = \{v\}$ is violated unless $\deg_{G-F}(v) \ge s+1$. (b) For $N = 4$, $\log^2 N = (\ln 4)^2 \approx 1.92$; the worst 2-subset has $|N(U)| = 1 \ge \varepsilon_0 \cdot 2/1.92$ iff $\varepsilon_0 \le 0.961 \ge 24/25$. (c) is immediate. □

Proof status: Complete proof in campaign records, `campaign/lane_12_expansion.md` Prop. 12.3(a), 12.3(b′), 12.3(c). The original part (b) claimed an asymptotic family of sparse expanders; it was withdrawn after a star-shaped counterexample and an unfixable accounting gap were found, and is replaced here by the finite statement (b) only..

Remark 10.4 (Optimality within the architecture). Within BM's architecture: the parameter s is monotone-constrained along the whole chain (its value $s = \log^{273} n$ being forced by the square of the expander-splitting threshold); the deletion price $4sn \log n$ is exact; the three-random-classes partition is essential (merging attempts die on closed walks that must avoid a forbidden vertex class); and by Propositions 10.1–10.3 the polylog remainder with tower reset cannot be beaten from inside. A formal intermediate rung $f(d) = 2 \uparrow\uparrow (\theta \log^* d)$ would give $O(n \log \log^* n)$, but no implementation path within the BM machine is known. We record these as architecture-relative findings.

11. Concluding remarks and open problems

The six-regular bottleneck that motivated Section 6 is resolved. What remains is a short list of problems, each with an identified obstruction.

Problem 11.1 (Planar graphs)**.** *Is $\mathrm{ce}(G) \le n-1$ for every planar G? Verified by exhaustive enumeration for $n \le 6$. The strengthening "$\mathrm{ce} \le n-2$ for all planar G containing a cycle" is false (K_4; and 1,273 counterexamples at $n = 6$), so any proof must exploit savings beyond triangles.*

Problem 11.2 (Torus)**.** *Is $\mathrm{ce}(G) \le n-1$ for every toroidal G? The bound, if true, is sharp ($\mathrm{ce}(K_{3,4}) = \mathrm{ce}(K_{3,5}) = \mathrm{ce}(K_{3,6}) = n-1$, Theorem 8.3) and would subsume the planar problem. Eulerian toroidal graphs are settled (Theorem 4.5).*

Problem 11.3 (Maximum degree five; general degree six)**.** *Does $\Delta \le 5 \Rightarrow \mathrm{ce} \le n-1$ (Problem 6.1 of [1])? And does $\Delta \le 6 \Rightarrow \mathrm{ce} \le n$? The even case of the latter is settled at $n-1$ (Corollary 6.10); what remains is the "long T-join" danger zone of Remark 6.13.*

Problem 11.4 (Below $n-2$ for six-regular)**.** *Can the two residual configurations of Theorem 6.7 — a six-regular graph containing two edge-disjoint triangle 2-factors, or one whose 4-regular remainder has no triangle 2-factor — be pushed to $\mathrm{ce} \le n-3$? Both require bypassing the per-2-factor ledger, which pins a triangle factor at m parts. The known truth is far lower ($\mathrm{ce} = 3$ at $n = 9$), so the question is about the method, not the constant.*

Problem 11.5 (A triangle-free 2-factor in every 4-regular graph)**.** *Does every 4-regular simple graph have a 2-factor with no triangle at all? The proof of Proposition 6.6 settles the line-graph case; computational reconnaissance on random 4-regular graphs ($n = 6..14$) and on the $L(B)$ family found no obstruction. The decision algorithm of Bertram–Horák [4] concerns the stronger two-factor version and gives no counting consequence; the characterisation results of Hartvigsen [18] for triangle-free 2-factors in general graphs should reposition this problem and was not yet integrated at the time of writing.*

Problem 11.6 (The $d = 3$ constant)**.** *Prove or refute Conjecture 8.4: is* $\sup\{\mathrm{ce}(G)/n : G \text{ is 3-degenerate}\} = 4/3$*, with extremal family* $K_{3,3k}$*? Equivalent to a* τ*-bound on* 3*-cores.*

Problem 11.7 (The 4-core)**.** *Is every graph whose* 4*-core is small "*$n-1$*-good"? Neither a proof nor a counterexample is known; the* $K_{\mathrm{odd},q}$ *mechanism needs odd degrees that* 4*-core candidates lack, making this the recommended next target of the degeneracy program.*

Problem 11.8 (Linear constants)**.** *Does* $f(n) \le \lceil 3n/2 \rceil + O(1)$*? Any explicit-constant* $O(n)$ *bound would be new. The bipartite version is governed by Hajós' conjecture: an upper bound* $\lfloor 3n/2 \rfloor - 1$ *for bipartite graphs is equivalent to the bipartite case of Hajós' cycle conjecture (1968) and is open; Hajós' conjecture has been verified for all Eulerian graphs on at most* 12 *vertices* [19]*. Known:* $f(n) \ge (3/2 - o(1))n$ [13, 6]*,* $f(n) \ge n$ *for* $n \ge 7$ *(Corollary 9.3).*

Problem 11.9 (Beyond Dirac regularity)**.** *For non-regular graphs with* $\delta \ge n/2$*, is* $\mathrm{ce} \le (1+\varepsilon)n$ *(or even* $\le n$*)? Hamilton decomposition fails off the regular path; the records locate the obstruction in the critical band. Similarly, for* 2*-connected* 4*-regular graphs, does* $\mathrm{ce} \le n/2 - 1$ *extend to all* $n \ge 10$*? (Proposition 9.4.)*

Problem 11.10 (Odd-regular graphs without perfect matchings, $k \ge 2$)**.** *Conjecture 6.4 of* [1] *is now settled for* $k = 1$ *(Theorem 7.6). For* $k \ge 2$ *the bridge route dies — a bridgeless* $(2k+1)$*-regular graph without a perfect matching is no longer excluded by Petersen — and the correct cut structure is that of Tutte sets; a piece lemma for near-regular blocks is the missing instrument. Computational support: Tutte-obstruction families of* 5*-regular graphs satisfy* $p \le n - 3$ *with large slack.*

Problem 11.11 (Tight instances)**.** *Characterise the graphs with* $\mathrm{ce}(G) = n-1$ *(Remark 9.5); even a structural dichotomy for* 2*-connected instances would sharpen the block ledger of Section 3.*

Problem 11.12 (Below $\log^*$)**.** *Find any machinery producing a decomposition of* $o(n \log^* n)$ *parts; formally the rung* $O(n \log\log^* n)$ *exists (Remark 10.4) but has no known implementation. Equivalently: beat the tower-reset dynamic of Proposition 10.1.*

Appendix A. Proof status and computational verification

Every theorem in this paper carries, after its statement or proof, a pointer to its complete proof in the project records. This appendix summarises the computational verification on which several of those records rely. Three principles governed the verification: every constructive step of a proof is implemented and checked by a validator independent of the construction logic (degrees, spanningness, cycle legality, partition completeness); every exhaustive claim is over a *complete* enumeration, not a sample; and every checker was subjected to negative controls (deliberately injected errors, all detected). Scripts are maintained with fixed random seeds; three full re-runs of the six-regular suite produced byte-identical certificates.

General results. This subsection covers Sections 3, 4, 8 and 9; the underlying records are the lane and notes files of the campaign archive.

- Exact anchors: $\mathrm{ce}(K_4) = 3$, $\mathrm{ce}(K_5) = 2$, $\mathrm{ce}(K_6) = 5$, $\mathrm{ce}(K_7) = 3$, $\mathrm{ce}(K_8) = 7$ (independent exact solvers; the K_8 value from a 3.3×10^7-step search log). $\mathrm{ce}(K_3 \cup K_{3,4}) = 7$, verified four independent ways (DP solver; branch-and-bound; feasibility enumeration at $k = 5, 6$; an independently written solver with zero shared code), plus per-edge certificate checking.
- Small planar values: $\mathrm{ce} \le n-1$ verified exhaustively for $n \le 6$; the $n-2$ strengthening refuted by 1,273 counterexamples at $n = 6$ (Problem 11.1).
- Degeneracy: 725 randomized 2-degenerate instances machine-checked for Theorem 8.1; 630 randomized 3-degenerate instances with exact ce and exact τ, zero violations of Conjecture 8.4.

- Machine certificates for Theorem 8.6: for $K_{3,14}+e$, all 3,324 cycles enumerated for the rate inequality, ce $\in [17,21]$; for $K_{5,11}$, all 794,530 cycles enumerated, ce ≥ 16; $\mathrm{ce}(K_{3,7})=10$ by exact solver.
- Exact small values for Theorem 9.1: $\mathrm{ce}(K_{2,5})=4$, $\mathrm{ce}(K_{3,4})=6$, $\mathrm{ce}(K_{3,7})=10$, $\mathrm{ce}(K_{5,5})=7$.

Six-regular graphs. This subsection covers Section 6; the records are `campaign/r2_01_*` and `campaign/harden_round2/`.

- Structural family, exhaustive: all marked cubic bipartite B with $s=3..6$ vertices per side — 1, 24, 2,040, 297,200, totalling 299,265 instances (counts match the theoretical enumeration exactly) — checked by both the constructive route and an independent brute-force route; in addition the Φ-mechanism was force-checked on every one of the 299,265 instances.
- Full parameter space: for $s\leq 5$, *all* B, *all* perfect matchings and *all* 2^s orientations — 832,944 combinations ($48+3{,}456+829{,}440$) — zero counterexamples; for $s=6$, 300 sampled B. (A two-round total of 1,183,920 combinations was accumulated across the two verification passes; the figures 829,440 and 832,944 refer to the $s=5$ single tier and the $s\leq 5$ total respectively.)
- Anti-Φ partition (Proposition 6.6): verified on the same 299,265 instances plus 660 random large B (s up to 40).
- Six-regular, $n=9$: all 30,016 labelled graphs (four isomorphism classes of sizes 280/5,040/4,536/20,160, matching the theoretical counts) — decomposition pieces $\leq 7=n-2$ throughout, the $n-2$ assertion enforced by an independent implementation; exact ce $=3$ for all four classes.
- Isomorphism-class exhaustive, $n=10/11/12$: 21, 266, 7,849 classes; maximum piece counts 8, 8, 10 against bounds $n-1$, $n-1$, $n-2$ — zero violations. For $n=10$, exact ce determined for every class by two independent engines (branch-and-bound and MILP), all equal to 3. At $n=12$, exactly 5 classes reach the ledger value $n-2=10$ (tight ledger witnesses; their exact ce is still 3).
- Labelled reconnaissance, $n=10$: 890,100 labelled graphs swept under a 600-second truncation (a sampling sweep, honestly not exhaustive; superseded for $n=10$ by the class-exhaustive item above).
- Random six-regular, $n=15/18/21/24$, 2,000 graphs each (8,000 total, multiple seeds): maximum piece counts 11/12/13/13 against bounds 13/16/19/22. The sampler (circulant start plus degree-preserving swap MCMC) yields correlated samples; these sweeps are treated as smoke tests, not as uniform-random evidence.
- Adversarial families at $s=7$ (1,500 instances) and $s=8$ (600 instances): partition and double triangle-freeness checks, all pass.
- Certificates: 18,279 decomposition certificates re-verified by an independent validator, zero failures; the validator passed 5/5 negative controls; three full re-runs produced sha256-identical outputs.
- Residual risk boundary, stated plainly: the structural danger zone of Lemma 6.5 is exactly the $L(B)$ family; the exhaustive coverage above reaches $s\leq 6$ (i.e. all potential counterexamples on $n\leq 18$ vertices), and the case $s\geq 7$ ($n\geq 21$) rests on the proof alone.

Cubic and odd-regular graphs. This subsection covers Section 7; the underlying records are the files of `front_tre/`.

- Exhaustive, connected odd-regular, $n\leq 12$ (degrees 3 through 11): 9,588 graphs, zero violations of the cubic/odd-regular claims; f_{re} exact values additionally determined for all instances with $m\leq 20$ (116 graphs).
- Exhaustive, connected cubic: $n=14$: 509 graphs; $n=16$: 4,060; $n=18$: 41,301 — zero violations of Theorem 7.6.

- Piece lemma census: all 60,077 connected bridgeless subcubic graphs on $n \leq 14$ vertices satisfy Lemma 7.4, with $\min(2p_2 - n) = 3$: the margin $n/2+3/2$ is attained, and attained *only* by $K_{2,3}$ (verified up to isomorphism). For even $n \leq 14$ the margin is at least 4.
- Perfect-matching-free census: minimal order 16, exactly one graph (structure: three subdivided-K_4 blades on one central vertex, $p = n/2 + 1$); at $n = 18$ exactly four graphs, all with the same blade structure; all perfect-matching-free cubic graphs found have $p = n/2 + 1$.
- 5-regular Tutte-obstruction families (orders 36 to 146, 5 graphs): $p \leq n - 3$ throughout.
- Cross-checks: p computed by two independent paths (brute-force pair enumeration vs. weighted matching), f_{re} by two paths (subset DP vs. exact-cover branch-and-bound), p_2 cross-checked by ILP; positive anchors (K_4, K_6 tight at $n - 1$; Petersen $= 6$; $C_5 = 1$; $P_4 = 3$) all recovered.

Acknowledgments

This work was carried out by a human-directed *multi-agent AI research system.* Independent "attack lane" agents proved the theorems in parallel from a shared briefing and were required to make every constructive claim independently recomputable (exact solvers, explicit certificates, exhaustive checks); an adversarial *verifier* agent then re-derived the proofs line by line, re-ran the verification scripts, and cross-checked quotations against local source copies of the cited papers. The first verification round issued a triage report: eleven results green-lit, eleven items flagged for repair (all at the level of argument, wording, or constants; none affected a theorem's conclusion), and one claim confirmed for withdrawal. A blue-team round implemented all repairs, including the replacement of an erroneous counting argument in Corollary 4.7, the narrowing of Proposition 10.1(b) to its tower-step form, the correction of an off-by-one in Remark 6.14 and of an arithmetic slip in the proof of Theorem 8.3, the correction of two numerical constants in Propositions 10.1–10.2, and the withdrawal of an asymptotic construction in Proposition 10.3, as well as of an earlier claim that a phase-transition argument yields $O(n \log\log\log n)$ — a claim whose direction of comparison was inverted, as Proposition 10.2 explains. A second, independent round re-reviewed the proof of the six-regular theorem (verdict: pass, with six minor findings, all repaired), widened the numerical coverage to the levels recorded in Appendix A, and re-examined the literature — where it surfaced the observation of Kouider–Sabidussi [20] and thereby forced the repositioning of Proposition 6.6 from a claimed-first to an explicit-special-case status, as well as the correction of the citation record of [1] to its second arXiv version. We believe this adversarial loop is itself a methodological contribution; the complete error ledger is maintained in the project repository.

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